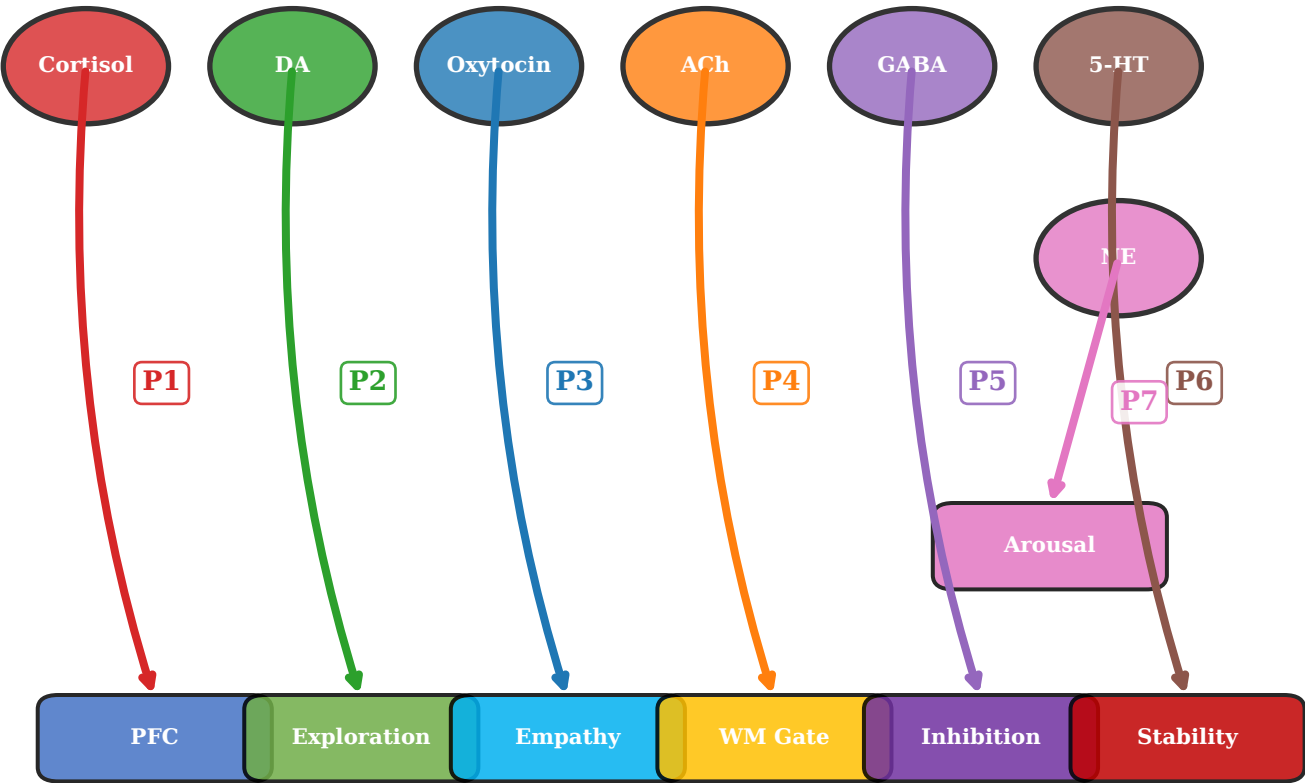


A: Coupling Pathways Network



Source nodes: Neurotransmitters

Target nodes: Cognitive functions

B: Pathway Mechanisms and Effects

ID	Pathway	Mechanism	Reference	Computational Effect
P1	Cortisol → PFC	Stress inhibits executive function	Arnsten 2009	$\text{PFC.inhibition} *= (1 - \text{cortisol} \times \alpha)$
P2	DA → Exploration	Dopamine drives exploration-exploitation	Schultz 2007	$\text{exploration_rate} = f(\text{DA}, \text{baseline})$
P3	Oxytocin → Empathy	Social bonding hormone	Dunbar 2009	$\text{resonance} *= (1 + \text{oxytocin} \times \gamma)$
P4	ACh → WM Gate	Attention modulates working memory	Hasselmo 1999	$\text{wm_update_rate} = \text{gate}(\text{ACh})$
P5	GABA → Inhibition	E/I balance control	GABA theory	$\text{activity} = \text{clamp}(x, \text{GABA_threshold})$
P6	5-HT → Stability	Serotonin promotes behavioral consistency	Dayan 2009	$\text{decision_variance} *= (1 - 5\text{-HT} \times \delta)$
P7	NE → Arousal	Adrenaline boosts salience	Aston-Jones 2005	$\text{attention_gain} = \text{SNR_boost}(\text{NE})$

Parameter Ranges (with Uncertainty)

P1 (α): 0.4 [0.2, 0.6] — Stress-PFC coupling
P2 (η): 0.2 [0.1, 0.4] — DA-exploration sensitivity
P3 (γ): 0.5 [0.2, 0.8] — Oxytocin-empathy modulation

Coupling Matrix A (lower triangular)

